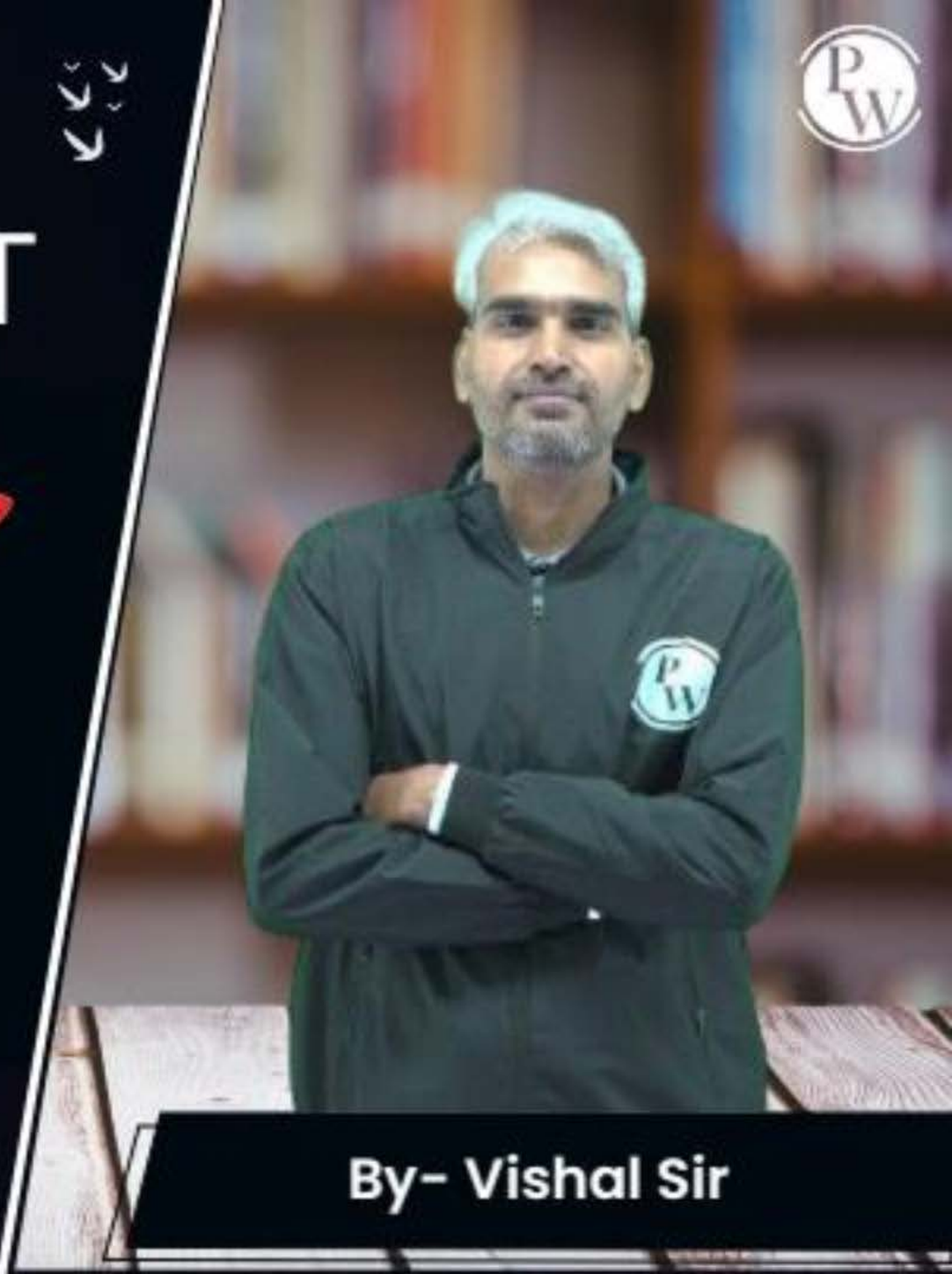
# Computer Science & IT



## Discrete Mathematics

## Mathematical Logic

**Lecture No. 06**



By- Vishal Sir

# Recap of Previous Lecture



**Topic** Quantifiers

**Topic** Scope of quanitifiers

Slide

# Topics to be Covered



Topic Negation of statement formula

Topic Sentences with multiple quantifiers

Topic Relationship diagram

Slide

H.W: Consider following two Predicate formula

$F_1$: $\forall x \{P(x) \longleftrightarrow Q(x)\} \longrightarrow \{\forall x \{P(x)\} \longleftrightarrow \forall x \{Q(x)\}\}$

$F_2$: $\{\forall x \{P(x)\} \longleftrightarrow \forall x \{Q(x)\}\} \longrightarrow \forall x \{P(x) \longleftrightarrow Q(x)\}$

Which of the following is true

(a) Only $F_1$ is valid
(b) Only $F_2$ is valid
(c) Both $F_1$ and $F_2$ are valid
(d) Neither $F_1$ nor $F_2$ is valid



# Topic : Note

- Let, P(x): x is true.
  ∴ ~P(x): x is not true / x is false

| | Form | Meaning |
|---|---|---|
| 1 | $\forall x\, P(x)$ | All are True / Not at least one false |
| 2 | $\exists x\, P(x)$ | At least one True / Not all are false |
| 3 | $\sim \forall x\, P(x)$ | Not all are True / At least one false |
| 4 | $\sim \exists x\, P(x)$ | Not at least one True / All are false |
| 5 | $\forall x \sim P(x)$ | All are false / Not at least one True |
| 6 | $\exists x \sim P(x)$ | At least one false / Not all are True |
| 7 | $\sim \forall x \sim P(x)$ | Not all are false / At least one false |
| 8 | $\sim \exists x \sim P(x)$ | Not at least one false / All are True |



# Topic : Equivalences

1. $\forall x\, P(x) \equiv \sim \exists x (\sim P(x))$

$\forall x (P(x)) \Leftarrow \forall x (\sim(\sim P(x)))$

2. $\exists x\, P(x) \equiv \sim \forall x (\sim P(x))$

$\exists x (P(x)) \Leftarrow \exists x (\sim(\sim P(x)))$

3. $\sim \forall x\, P(x) \equiv \exists x (\sim P(x))$

$\exists x (\sim P(x)) \equiv \exists x (\sim P(x))$

4. $\sim \exists x\, P(x) \equiv \forall x (\sim P(x))$

$\forall x (\sim P(x)) \equiv \forall x (\sim P(x))$



# Topic : Negation of a statement formula

When negation sign is propagated inside the Predicate formula,

then
- Replace $\forall$ by $\exists$
  and
  replace $\exists$ by $\forall$
  and
- Negate the scope of the corresponding quantifier

Q: Negate the predicate formula $\forall x \{ P(x) \rightarrow S(x) \}$

Solution:

Negation of $\forall x \{ P(x) \rightarrow S(x) \}$

$$\equiv \sim \{ \forall x \{ P(x) \rightarrow S(x) \} \}$$

$$\equiv \exists x \{ \sim ( P(x) \rightarrow S(x) ) \}$$

$$\equiv \exists x \{ \sim ( \sim P(x) \vee S(x) ) \}$$

$$\equiv \boxed{\exists x \{ P(x) \wedge \sim S(x) \}}$$

Q: The predicate formula $\forall x \{ B(x) \wedge I(x) \}$ is equivalent to which of the following predicate formula

(a) $\exists x \{ B(x) \rightarrow \sim I(x) \}$

(b) $\sim \exists x \{ B(x) \rightarrow \sim I(x) \}$

(c) $\sim \exists x \{ \sim B(x) \rightarrow I(x) \}$

(d) $\exists x \{ \sim B(x) \rightarrow I(x) \}$

$$A \equiv \sim(\sim A)$$

$$\forall x \{ B(x) \wedge I(x) \} \equiv \sim(\sim(\forall x \{ B(x) \wedge I(x) \}))$$

$$\equiv \sim \{ \exists x (\sim B(x) \vee \sim I(x)) \}$$

$$\equiv \sim \exists x (B(x) \rightarrow \sim I(x))$$

$$\equiv \sim \exists x (I(x) \rightarrow \sim B(x))$$

## Sentence with multiple quantifiers :→

P : "likes" , Domain = Set of all students in our class

$P(x, y)$ = $x$ likes $y$ ≡ $y$ is liked by $x$

① $\forall x \forall y \{P(x,y)\}$ : All $x$ likes All $y$

Every student in our class likes Every student of class

② $\forall y \forall x \{P(x,y)\}$ : All $y$ liked by all $x$

Every student is liked by Every student

$$\forall x \forall y \{P(x,y)\} \equiv \forall y \forall x \{P(x,y)\}$$

## Sentence with multiple quantifiers :→

P : "likes" , Domain = Set of all students in our class

$P(x,y)$ = $x$ likes $y$ ≡ $y$ is liked by $x$

③ $\exists_x \forall_y \{P(x,y)\}$ : At least one $x$ likes all $y$
Someone in our class likes Everyone

④ $\forall_y \exists_x \{P(x,y)\}$ : All $y$ liked by at least one $x$
Everyone is liked by some one

$\exists_x \forall_y \{P(x,y)\} \Longrightarrow \forall_y \exists_x \{P(x,y)\}$ is Valid

$\not\Longleftarrow$ is invalid

## Sentence with multiple quantifiers :→

P : "likes" , Domain = Set of all students in our class

$P(x, y)$ = $x$ likes $y$ ≡ $y$ is liked by $x$

⑤ $\exists y \, \forall x \, \{P(x,y)\}$ : Someone is liked by Everyone

At least one 'y' is liked by all x

At least one student in our class is liked by Every student of class

⑥ $\forall x \, \exists y \, \{P(x,y)\}$ : Everyone likes Somebody

All x likes some y

Every student of class likes some one from the class

$\exists y \, \forall x \, \{P(x,y)\} \Longrightarrow \forall x \, \exists y \, \{P(x,y)\}$ is valid

$\not\Longleftarrow$ need not be true

## Sentence with multiple quantifiers :→

P : "likes" , Domain = Set of all students in our class

$P(x, y)$ = $x$ likes $y$ ≡ $y$ is liked by $x$

⑦ $\exists x \exists y \{P(x,y)\}$ : Some one likes Some one

⑧ $\exists y \exists x \{P(x,y)\}$ : Some one is liked by Some one

$$\exists x \exists y \{P(x,y)\} \equiv \exists y \exists x \{P(x,y)\}$$

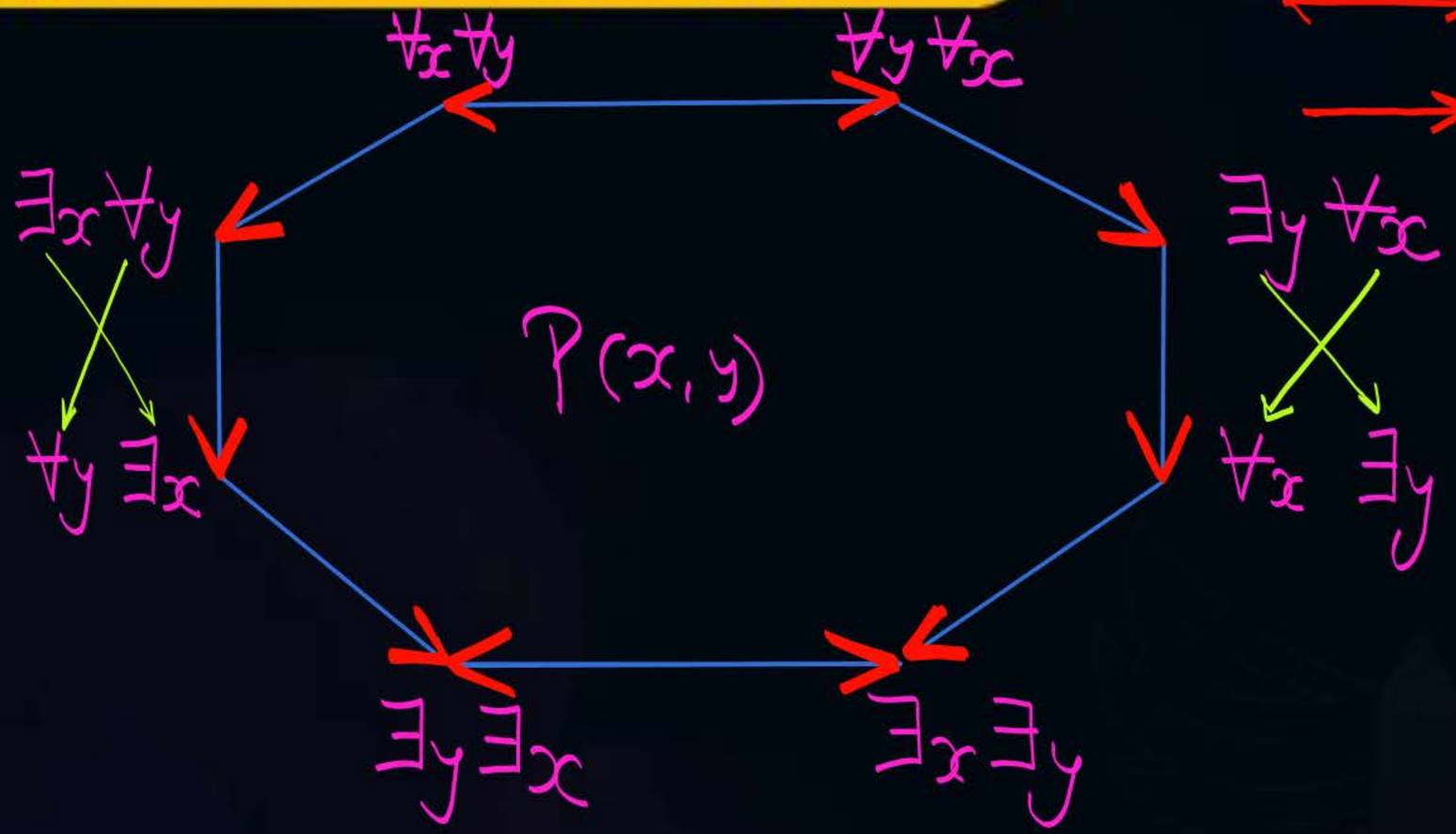
① $\forall x \forall y \{P(x,y)\} \equiv \forall y \forall x \{P(x,y)\}$

② $\exists x \forall y \{P(x,y)\} \not\equiv \forall y \exists x \{P(x,y)\}$

③ $\exists y \forall x \{P(x,y)\} \not\equiv \forall x \exists y \{P(x,y)\}$

④ $\exists x \exists y \{P(x,y)\} \equiv \exists y \exists x \{P(x,y)\}$

# Topic : Relationship diagram




$\longleftrightarrow$ represent Equivalence

$\longrightarrow$ represent implication

[MSQ]

#Q. Which of the following is/are true?

A $\forall x \exists y\ P(x,y) \nRightarrow \exists y \forall x P(x,y)$

B $\forall x \forall y\ P(x,y) \equiv \exists y \forall x P(x,y)$

C $\exists x \forall y\ P(x,y) \Rightarrow \forall y \exists x\ P(x,y)$

D $\exists x \exists y\ P(x,y) \nRightarrow \forall y \exists x\ P(x,y)$

[MCQ]

#Q. Consider the first order logic sentence $F: \forall x(\exists y\ R(x,y))$. Assuming non-empty logical domains, which of the sentences below are implied by $F$?

I. $\exists y\ (\exists x\ R(x,y))$

II. $\exists y\ (\forall x\ R(x,y))$

III. $\forall y(\exists x\ R(x,y)$

IV. $\sim\exists x(\forall y\sim R(x,y))$

$\forall x\ \exists y\ R(x,y) \equiv F$

A IV only

B I and IV only

C II only

D II and III only



# 2 mins Summary

Topic Negation of statement formula

Topic Sentences with multiple quantifiers

Topic Relationship diagram

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# THANK - YOU

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